LEGALKART PAPER GENERATOR

LegalKart Paper Generator — Product & Technical Specification

Product and technical specification: tech stack, database schema, integration flows, template JSON Schema, build plan and sample code.

Version 1.0 · October 2026 · India

Version: 1.0 · **Date:** 16 Oct 2026 · **Market:** India

What it does: anyone can produce a ready-to-sign property document (a rent agreement, an agreement to sell, an affidavit and so on) by entering the parties' details. The app:

1. Maps the property pincode to its state, district, Sub-Registrar Office and stamp duty.
2. Verifies tenants and buyers by OTP.
3. Buys the correct e-stamp online.
4. Outputs a stamped PDF and an editable DOCX.

> **Legal disclaimer:** all stamp-duty figures in this repository are **illustrative placeholders**. Rates must be taken from state notifications and signed off by an advocate before launch (see Compliance).

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0. Repository map

Path	What it is
`SPEC.md`	This document
`db/schema.sql`	PostgreSQL 16 schema. Includes the 4 new tables: `pincodes`, `stamp_rates`, `e_stamp_transactions`, `party_verifications`. Tested by loading into Postgres 16
`db/seed.sql`	Development seed (DL, HR, KA, MH). The rates in it are illustrative
`schemas/document-template.schema.json`	JSON Schema for templates: form, parties, OTP policy, pincode binding, duty inputs, e-stamp mapping, clauses, output
`schemas/document-instance.schema.json`	JSON Schema for a filled document: verification status, `pincodeData`, `dutyQuote`, `eStamp`

Path	What it is
`templates/rent-agreement.residential.json`	A production-shaped Rent Agreement template (DL/HR/KA)
`server/`	Reference backend in TypeScript/Fastify: pincode resolver, duty engine, OTP, e-stamp adapters, Razorpay, PDF/DOCX. Has 13 tests
`mobile/RentAgreementCheckout.tsx`	React Native screens: party OTP card and e-stamp checkout
`samples/`	Output of `npm run demo`: stamped PDF (sandbox certificate), DOCX, HTML and an instance JSON

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1. Product scope

1.1 Users

Persona	Need
Landlord / owner (primary payer in the MVP)	A valid rent agreement in 10 minutes, with no trip to a stamp vendor
Tenant	Proof of address, a basis for HRA claims, and assurance that the paper is genuine. Verifies by OTP
Buyer / seller	A draft agreement to sell or a token-money receipt, on correct-value stamp paper
Broker / property manager (Phase 2)	Many agreements a month: saved parties, bulk generation, a wallet
Ops agent (internal)	Fulfils manual e-stamps, curates pincode-to-SRO mappings, reconciles payments
Advocate partner (Phase 2)	Paid review of a draft before signing

1.2 Document catalogue

The brief refers to a "previous list" that isn't available in this context, so the catalogue below is a reconstruction for Indian property transactions. Adjust it if the original list differs.

#	Document	Category	E-stamp	Registration	Mandatory OTP roles	Aadhaar e-sign allowed?[1]	Phase
1	Residential Rent Agreement (≤ 11 months)	Rental	✔	Optional (compulsory if > 11 months)	Tenant	✘ (counsel to confirm)	**MVP**
2	Leave & Licence Agreement (Maharashtra)	Rental	✔	**Compulsory**, any term (s.55 MRC Act)	Licensee	✘	P2

#	Document	Category	E-stamp	Registration	Mandatory OTP roles	Aadhaar e-sign allowed?[1]	Phase
3	Commercial Lease Deed	Rental	✔	If > 11 months	Tenant	✘	P2
4	Rent Receipt	Rental	Revenue stamp if cash > ₹5,000	–	–	✔	**MVP**
5	Lease Termination / Surrender Letter	Rental	Varies	–	Tenant	✔	P2
6	Agreement to Sell	Sale	✔	State-dependent[2]	Buyer	✘	**MVP** (draft + e-stamp)
7	Sale Deed (Conveyance)	Sale	✔	**Compulsory**	Buyer	✘	P3 (prepare + book SRO slot)
8	Token / Booking Receipt (Bayana)	Sale	✔ (state-specific)	–	Buyer	✘	P2
9	Possession Letter	Sale	Varies	–	Buyer	✔	P2
10	Gift Deed	Transfer	✔	**Compulsory**	Donee	✘	P3
11	Release / Relinquishment Deed	Transfer	✔	**Compulsory**	Releasee	✘	P3
12	Partition Deed / Family Settlement Memo	Transfer	✔	Compulsory / varies	All	✘	P3
13	General Power of Attorney	Authority	✔	Compulsory for immovable-property sale in many states	Attorney	✘ (IT Act excluded)	P2
14	Special Power of Attorney	Authority	✔	Varies	Attorney	✘	P2
15	Revocation of POA	Authority	✔	If original registered	–	✘	P3
16	Mortgage Deed / Memorandum of Deposit of Title Deeds	Finance	✔	Varies	–	✘	P3

#	Document	Category	E-stamp	Registration	Mandatory OTP roles	Aadhaar e-sign allowed?[1]	Phase
17	General Affidavit (address, name, income)	Declaration	✔ (small fixed)	– (notary / oath commissioner)	Deponent (self)	✘	**MVP**
18	Indemnity Bond	Declaration	✔	–	–	✔	P2
19	NOC (society or owner, for electricity, GST or passport)	Declaration	Plain paper or small stamp	–	–	✔	**MVP**
20	Will	Declaration	No stamp duty	Optional	–	✘ (IT Act excluded)	P3

[1] Under the IT Act 2000, First Schedule, electronic signatures are **not valid** for a power of attorney, a trust, a will, or "any contract for sale or conveyance of immovable property or any interest in such property". A lease is a transfer of an interest in immovable property, so leases default to **wet-ink signatures** until counsel gives an opinion. [2] Some states (e.g. Maharashtra) require an agreement to sell that involves possession to be stamped like a conveyance.

1.3 Feature requirements (mapped to the brief)

#	Requirement	How it is met
1	Smart dynamic forms with per-document schemas	Each template defines `sections[].fields[]` with types, validation, `showIf` conditions and repeaters. The app renders the form from the JSON. The server re-validates with the same rules (`validateForm`)
2	Profile with auto-fill	`user_profiles` plus `saved_contacts` . A field's `autofillFrom: "profile.fullName"` pulls from the profile, and "This is me" fills a whole party from the profile
3	Multi-party management with OTP for tenants and buyers	`document_parties` enforces min/max per role from the template. Roles in `verification.requiredRoles` get `otp_required = true` , and the quote is **blocked** until they verify. Changing a party's phone resets verification
4	State and pincode selection	Pincode → candidate post offices, districts and SROs. Auto-selects when one primary SRO has confidence ≥ 80; otherwise the user picks. The result drives `stamp_rates` lookup and the registration fee
5	E-stamp purchase	Pay (Razorpay) → `EStampService` → provider adapter (SHCIL partner, state GRAS, aggregator, or manual ops) → certificate number stored → printed on every page
6	Templates in legal English, state-compliant	Clauses can be state-scoped (`states: ["DL","HR"]`). Templates are versioned and go through `LEGAL_REVIEW` before `PUBLISHED`
7	Output: stamped PDF and editable DOCX	Puppeteer/Playwright renders HTML to PDF; pdf-lib puts the certificate first and adds a footer with the certificate number, page *x* of *y* and a verification QR; `docx` builds the Word file with signature blocks

#	Requirement	How it is met
8	Disclaimer and lawyer review	The user must acknowledge the disclaimer before paying (`consents` row). The "Get it reviewed by an advocate" upsell arrives in Phase 2

1.4 End-to-end user journey (Rent Agreement)

```
Home → Choose "Rent Agreement" → Property pincode → [confirm SRO if ambiguous]
     → Property & rent form (auto-filled from profile where possible)
     → Add landlord(s), tenant(s), witnesses
     → Tenant OTP (SMS / WhatsApp)  ── blocked until verified
     → Draft preview (watermarked "DRAFT — NOT STAMPED")
     → Duty quote + fees + GST → acknowledge disclaimer → Pay (UPI/card/netbanking)
     → E-stamp issued (cert no.) → Stamped PDF + DOCX → Print, sign in ink, (notarise)
```

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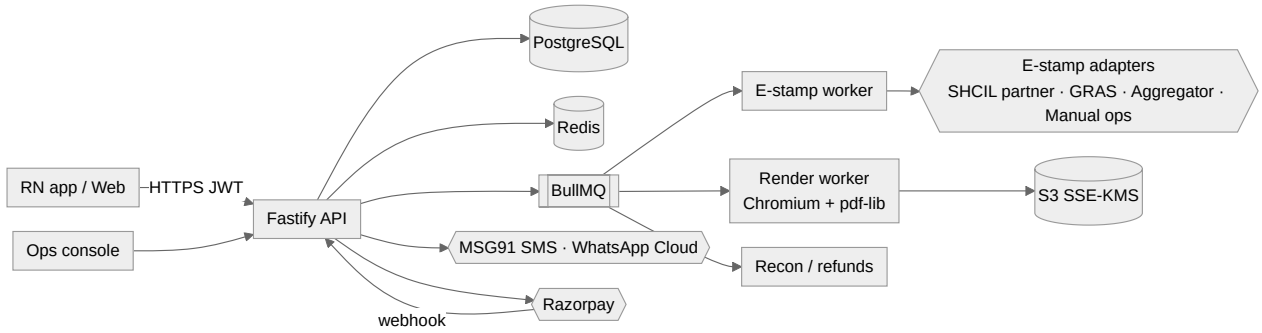
2. (a) Tech stack

Layer	Choice	Why / notes
Mobile app	React Native 0.8x + Expo (EAS Build), TypeScript, Expo Router	One codebase for iOS and Android. Shares zod types and form logic with the backend. OTA updates for template-driven UI
Forms	react-hook-form + zod, generated from the template JSON	Dynamic per-document schemas
State and data	TanStack Query, Zustand	Server cache; offline draft persistence
OTP autofill	iOS `textContentType="oneTimeCode"`; Android SMS Retriever (`react-native-otp-verify` , 11-char app hash in the DLT template)	No manual typing when the party is on the same phone
Payments SDK	`react-native-razorpay` (UPI intent, cards, netbanking)	
PDF viewer	`react-native-pdf`	In-app preview
Secure storage	`expo-secure-store` (Keychain / Keystore)	Refresh token
Web (optional)	React + Vite (consistent with this repo), same API	Desktop users and brokers
Admin / ops console	React + Vite + TanStack Table, behind Google SSO	Manual e-stamps, SRO curation, rate editor with 4-eyes approval
Backend API	Node.js 22 LTS, TypeScript, **Fastify 5**, zod	Fast, schema-first. The sample in `server/` uses this stack
ORM / SQL	Drizzle ORM or Kysely over `pg`	Typed SQL; migrations via drizzle-kit

Layer	Choice	Why / notes
Jobs	**BullMQ** on Redis	E-stamp purchase, status polling, PDF rendering, refunds, reconciliation, rate-change alerts
Database	**PostgreSQL 16** (AWS RDS, Mumbai `ap-south-1`, Multi-AZ) with `pg_trgm`, `btree_gist`, `citext`	Exclusion constraint prevents overlapping stamp-rate rows
Cache / rate limits	Redis 7 (ElastiCache)	OTP throttles, pincode cache, idempotency locks
Object storage	S3 (`ap-south-1`) with SSE-KMS, Object Lock on issued documents, short-lived presigned URLs	Data localisation and tamper evidence
Document rendering	Handlebars → HTML → **headless Chromium** (Playwright/Puppeteer) → PDF; **pdf-lib** to merge and stamp; **qrcode**; **docx** for Word	Pixel-identical PDF and editable DOCX from one model. Noto Serif / Noto Sans Devanagari fonts for regional languages
Payment gateway	**Razorpay** (primary), **Cashfree** or **PayU** (fallback)	UPI, cards, netbanking. Orders API with webhooks (`payment.captured`, `refund.processed`) and the Refunds API. Government duty and the taxable fee are accounted separately
SMS gateway	**MSG91** (primary), Gupshup or Kaleyra (fallback). All via **TRAI DLT** (entity ID, sender header e.g. `LGLKRT`, content template ID)	DLT is mandatory for commercial SMS in India
WhatsApp OTP	**WhatsApp Business Cloud API** (Meta), or a BSP such as Gupshup or Interakt; *authentication* category template with a copy-code button	Higher delivery rates; falls back to SMS
E-stamp	Adapter layer: SHCIL partner/ACC integration, state GRAS portals (Maharashtra GRAS, Haryana e-GRAS), licensed aggregators (e.g. stamp2india-type vendors, under contract), and a **manual-ops** fallback	See §4.1. There is no open public SHCIL purchase API
Pincode database	**India Post All-India Pincode Directory** (data.gov.in, OGD licence) + **LGD** codes (Local Government Directory) + a curated pincode → SRO table from state IGR portals	Refreshed monthly. Unofficial public pincode APIs are used only as a fallback lookup, never as the source of truth
Identity (Phase 2)	DigiLocker (issued documents), PAN verification (via a KYC provider), **Aadhaar eSign** through a licensed ESP (e.g. eMudhra, Protean/NSDL e-Gov, C-DAC) for documents where e-sign is permitted	
Auth	Phone OTP login (same OTP service), JWT access token (15 min) + rotating refresh token (30 days)	
Infra	AWS Mumbai: ECS Fargate (API, workers, renderer), ALB + WAF, CloudFront, Secrets Manager, KMS; Terraform	Data stays in India (DPDP Act and RBI expectations)

Layer	Choice	Why / notes
Observability	OpenTelemetry → Grafana/Tempo/Loki, Sentry (app and API), PagerDuty	SLOs on OTP delivery, e-stamp latency and PDF generation
CI/CD	GitHub Actions (typecheck, tests, schema validation of every template, SQL migration dry-run), EAS for mobile	
Analytics	PostHog (self-hosted in India)	Funnel: pincode → form → OTP → pay → stamped

2.1 Architecture



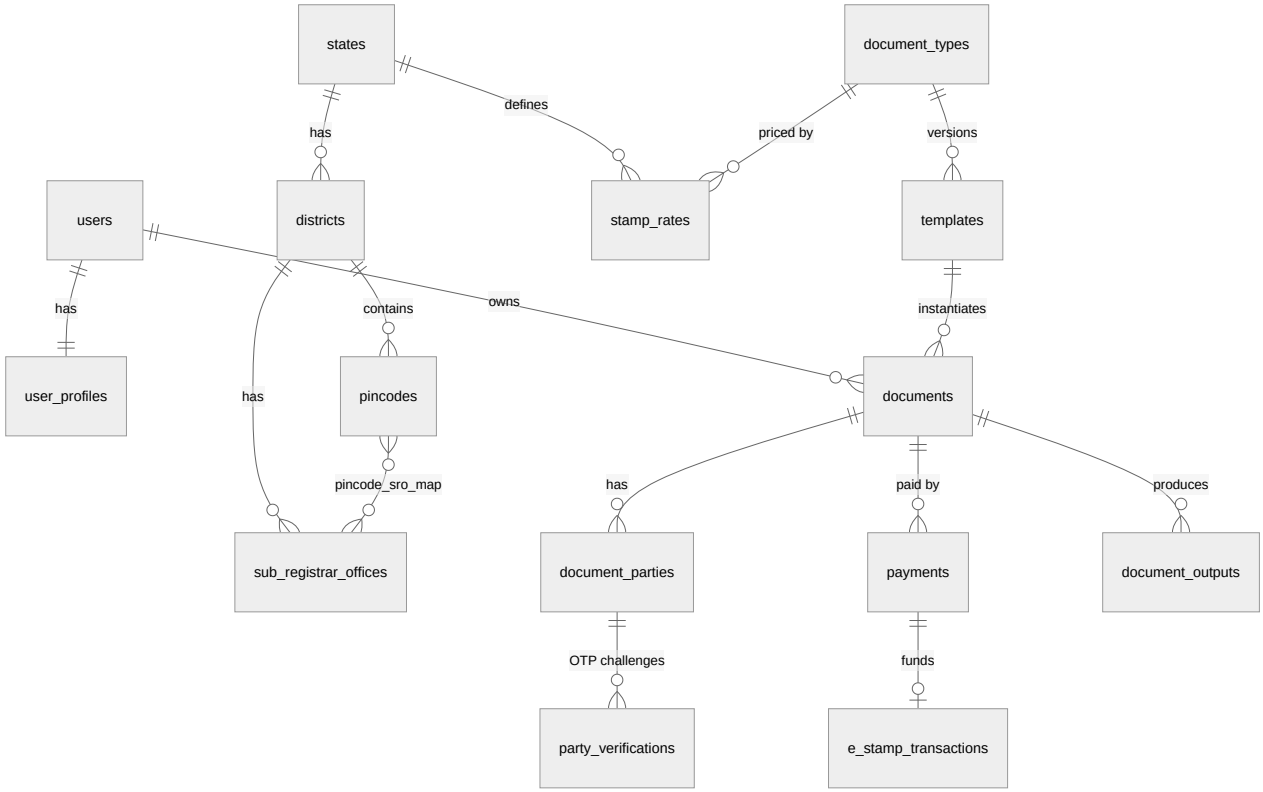
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3. (b) Database schema

The full DDL is in `db/schema.sql` and has been tested on PostgreSQL 16. Conventions:

- Money is stored as **paise in `BIGINT`**.
- Rates are stored in **basis points**.
- PII is envelope-encrypted (`*_enc`).
- **Full Aadhaar numbers are never stored**; only the last 4 digits.

3.1 Entity map



3.2 The four new tables

`pincodes` : one row per (pincode, post office). Seeded from the India Post directory.

Column	Notes
`pincode CHAR(6)`	`CHECK ~ '^[1-9][0-9]{5}$'`
`office_name`, `office_type` (HO/SO/BO), `delivery`	From India Post
`district_id → districts`, `state_code → states`	Joined to LGD codes
`area_type`	Curated (URBAN / RURAL / METRO …). Drives area-specific duty
`latitude`, `longitude`, `source`, `source_updated_on`	For freshness tracking

Companion table `pincode_sro_map(pincode, sro_id, is_primary, confidence)`. A partial unique index allows only one primary per pincode. Confidence below 80 makes the app ask the user to confirm.

`stamp_rates` : data-driven duty and fee rules. Rates live here, never in code or templates.

Column	Notes
`state_code`, `district_id` (NULL = state-wide), `area_type`	Scope
`document_type`, `charge_type`	`STAMP_DUTY`, `REGISTRATION_FEE`, `SURCHARGE`, `CESS` …
`basis`	`FIXED`, `CONSIDERATION`, `MAX_CONSIDERATION_MARKET_VALUE`, `AVG_ANNUAL_RENT_PLUS_DEPOSIT`, `TOTAL_RENT_PLUS_DEPOSIT`, `MH_LEAVE_AND_LICENCE`, `SLAB`
`rate_bps`, `fixed_amount_paise`, `min_amount_paise`, `max_amount_paise`, `slabs JSONB`	Amounts
`conditions JSONB`	e.g. `{"buyerGender":"FEMALE"}`, `{"maxTermMonths":11}`
`rounding_unit_paise`, `priority`	Rounding (always up) and tie-break
`effective_from`, `effective_to`	**Exclusion constraint** so no two rows for the same scope and conditions overlap in time
`source_ref`, `verified_by`, `verified_at`	Legal provenance. Production refuses to quote from an unverified row

`e_stamp_transactions`: one per purchase.

Column	Notes
`idempotency_key UNIQUE`	`estamp:<documentId>:<paymentId>`. Sent to the vendor; retries reuse it
`provider`, `provider_request_id`, `status`	State machine in §4.1
`fulfilment_mode`	`DIGITAL_SELF_PRINT` / `ACC_PRINT_COURIER` / `ACC_PICKUP` / `FRANKING`
`article_code`, `description_of_document`, `property_description`, `consideration_paise`, `stamp_duty_paise`, `first_party_name`, `second_party_name`, `duty_paid_by`, `purchased_by`	Exactly the fields printed on an e-stamp certificate
`certificate_no UNIQUE`, `unique_doc_reference`, `issued_at`, `acc_code`, `certificate_file_key`, `verification_url`	Issued certificate. `CHECK (status <> 'ISSUED' OR certificate_no IS NOT NULL)`
`raw_request`, `raw_response` (redacted), `attempt_count`, `next_poll_at`, `failure_reason`	Audit and polling

A partial unique index allows **only one active stamp per document**.

`party_verifications`: one row per OTP challenge.

Column	Notes
`party_id`, `document_id`, `phone_e164`, `channel`	

Column	Notes
`otp_hmac`	HMAC-SHA256(pepper, `id:otp`). **The plaintext OTP is never stored**
`attempts`, `max_attempts`, `resend_count`, `status`	`PENDING → SENT → VERIFIED / LOCKED / EXPIRED / FAILED`
`provider`, `provider_message_id`, `dlt_template_id`, `delivery_status`	Delivery receipts (DLR)
`expires_at`, `verified_at`, `consent_text_version`, `request_ip`, `user_agent`	Evidence that the party consented

A partial unique index allows **one live challenge per party**.

3.3 Key queries

```
-- Pincode → jurisdiction
SELECT p.office_name, p.area_type, d.name AS district, s.code, s.name, s.estamp_enabled
FROM pincodes p JOIN districts d ON d.id = p.district_id JOIN states s ON s.code = p.state_code
WHERE p.pincode = $1;

-- Candidate rate rows (the engine then picks the most specific in code)
SELECT * FROM stamp_rates
WHERE state_code = $1 AND document_type = $2
  AND effective_from <= $3 AND (effective_to IS NULL OR effective_to >= $3)
  AND (district_id IS NULL OR district_id = $4)
  AND area_type IN ('ANY', $5);

-- Stuck e-stamps for the poller
SELECT id FROM e_stamp_transactions
WHERE status IN ('SUBMITTED','MANUAL_REVIEW') AND next_poll_at <= now()
ORDER BY next_poll_at LIMIT 100 FOR UPDATE SKIP LOCKED;
```

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4. (c) Integration flows

4.1 E-stamp purchase (SHCIL / StockHolding / state portals / aggregators)

Market reality. *SHCIL* and *StockHolding* are the same entity: Stock Holding Corporation of India Ltd. It is the **Central Record Keeping Agency (CRA)** for e-stamping in most states that use e-stamps. There is **no open, self-serve public purchase API**, so integration is commercial. Options, in order of preference per state:

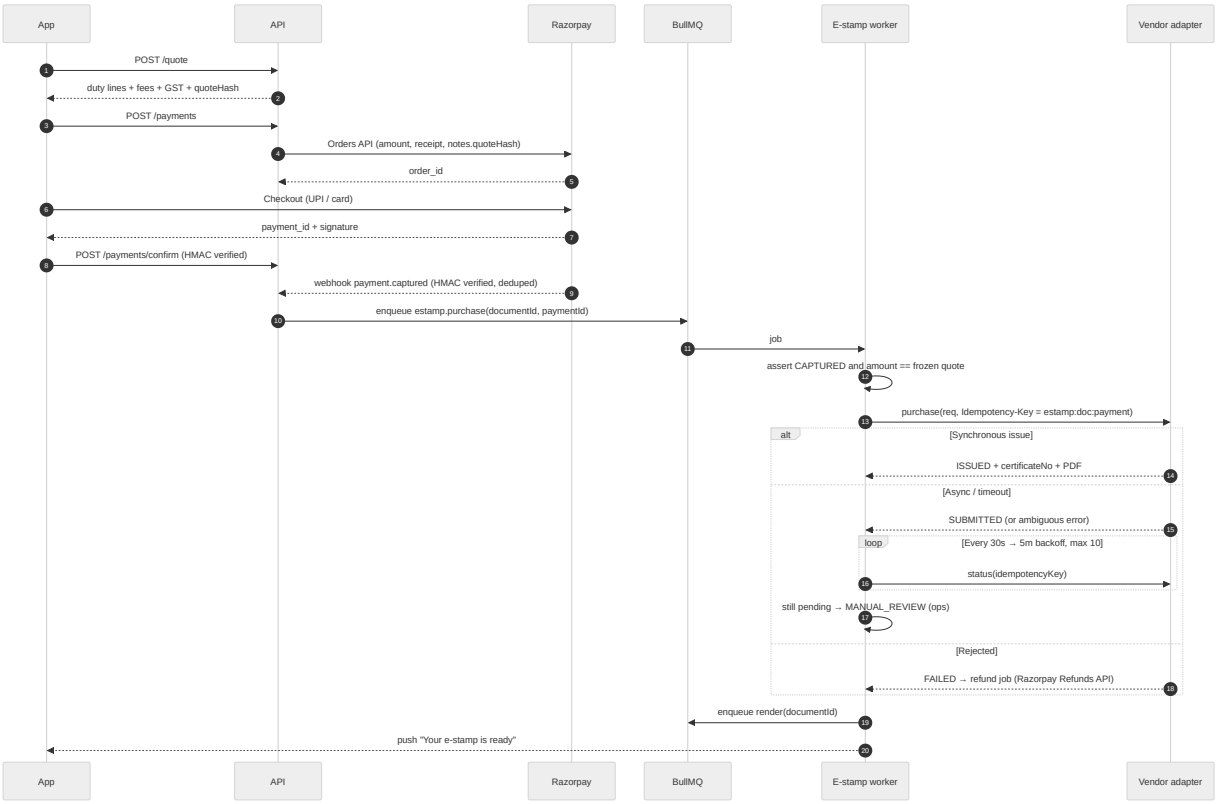
Route	How	Typical use
SHCIL partner / ACC	Register as (or partner with) an Authorised Collection Centre. Use the API or partner interface SHCIL provides under agreement	DL, KA, and other SHCIL e-stamp states

Route	How	Typical use
State treasury / GRAS	Pay duty through the state's government receipt portal (e.g. **Maharashtra GRAS** e-SBTR, **Haryana e-GRAS** e-stamp) under that state's rules	MH, HR and similar
Licensed aggregator	A contracted vendor (e.g. the *stamp2india*-type providers in the brief) that buys through its own ACC network and returns the certificate	Fast multi-state coverage at launch
Manual ops (always on)	The order is queued. An ops agent buys at a partner ACC and uploads the scan within an SLA	States with no API, and as a fallback
Franking (Phase 3)	Through authorised franking banks	High-value deeds

Certificate fields. These map 1:1 to columns in `e_stamp_transactions`:

- Certificate No. (e.g. `IN-DL12345678901234X`)
- Issued date
- Account reference (ACC)
- Unique Doc. Reference
- Purchased by
- Description of document (article)
- Property description
- Consideration price
- First party
- Second party
- Stamp duty paid by
- Stamp duty amount

Fulfilment. In some states the certificate must be printed by the ACC on security paper. In others an online, self-printable certificate is allowed. The template's `fulfilmentModes` together with state config decide between `DIGITAL_SELF_PRINT` and `ACC_PRINT_COURIER`. When couriering, the stamped PDF is still produced immediately, with the certificate number in the footer, and the user attaches the original certificate on arrival.



State machine:

```
INITIATED ─► SUBMITTED ─► ISSUED ─► (FULFILMENT_PENDING ─► DELIVERED)
     │            │   └──► MANUAL_REVIEW ─► ISSUED (ops upload) / FAILED
     └──► FAILED ─► REFUND_REQUESTED ─► REFUNDED
```

Safety rules (implemented in `server/src/estamp/estampService.ts`):

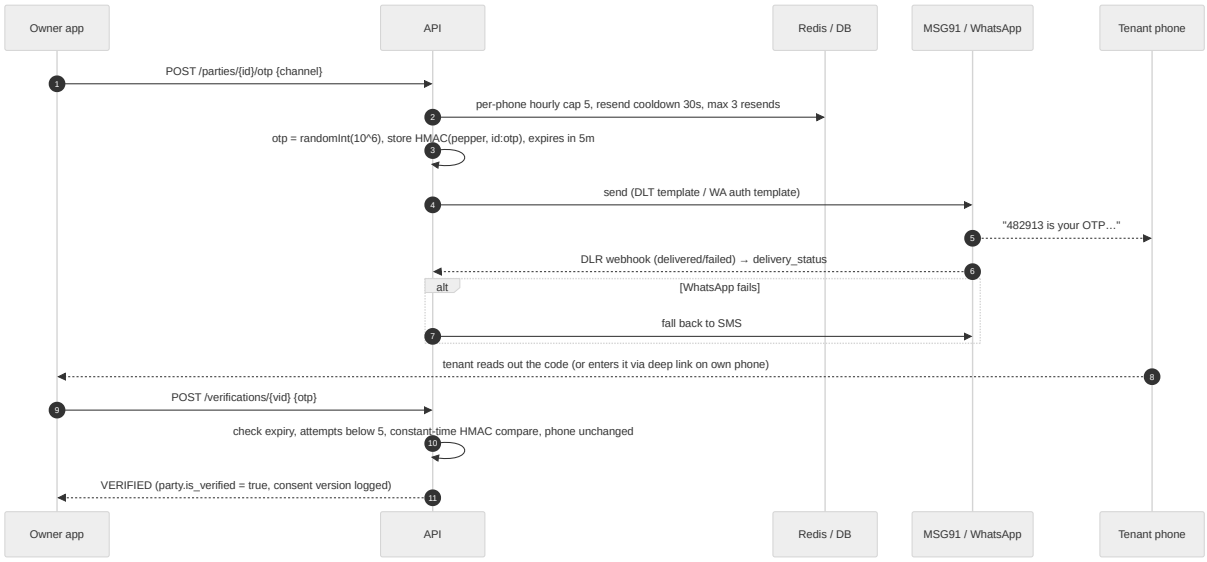
1. A purchase happens only after the payment is `CAPTURED`, and only for the **exact frozen quote**. `payments.duty_quote_hash` must equal `documents.duty_quote_hash`.
2. There is **one idempotency key per (document, payment)**. Webhook replays and app retries can never buy twice. The partial unique index enforces this in the database too.
3. An **ambiguous error** (a timeout or 5xx after sending) is never retried with a new key. The transaction moves to `SUBMITTED` and is resolved by status lookup on the same key.
4. A definite rejection → `FAILED` → automatic refund of the full amount. The refund window for unused or spoiled stamps follows the state's Stamp Act provisions; ops handle vendor-side cancellation.
5. Daily **reconciliation** checks Razorpay settlements against `payments`, and vendor statements against `e_stamp_transactions`. Mismatches open ops tickets.

Money flow. Stamp duty is government money collected as a pure pass-through. It is shown as a separate line, carries no GST, and is settled to the vendor or ACC float (Razorpay Route or a separate settlement account; confirm the structure with the PA and an accountant). LegalKart's service fee and the vendor's charge carry **18% GST**, and a tax invoice is issued.

4.2 SMS / WhatsApp OTP verification

Pre-requisites:

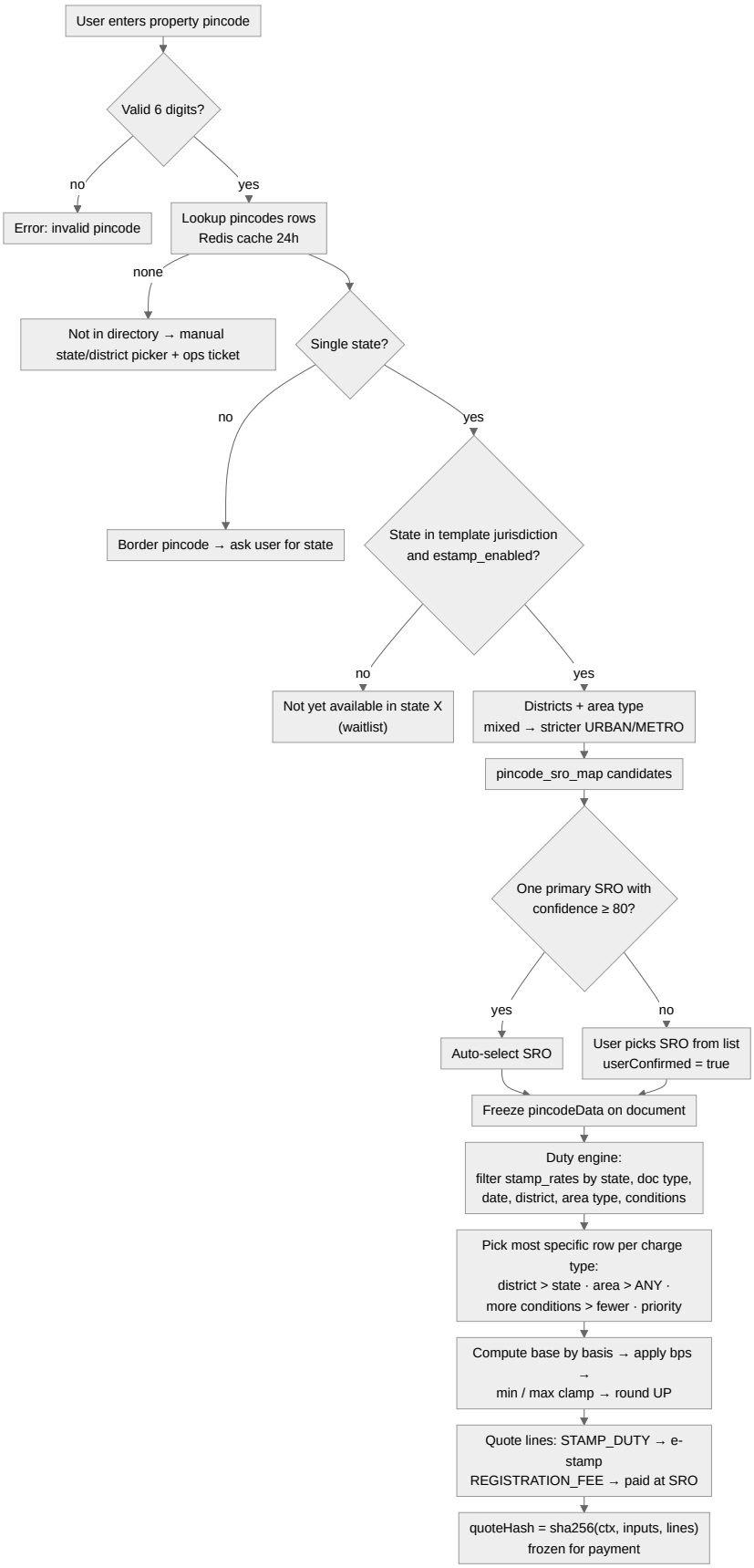
- **TRAI DLT** registration: principal entity, sender header (e.g. `LGLKRT`) and a content template such as: `{#var#} is your OTP to confirm your mobile number for a {#var#} on LegalKart. Valid for {#var#} minutes. Do not share it with anyone. -LGLKRT {#var#}` (The last variable is the Android SMS Retriever app hash.)
- A **WhatsApp authentication template** with a copy-code button, approved by Meta.



Rules (in `server/src/otp/otpService.ts`):

- The code is 6 digits from a CSPRNG, valid for 5 minutes.
- After **5 wrong attempts** the challenge is `LOCKED`.
- Only one live challenge exists per party; a new send supersedes the old one.
- **If the phone number changes, the party must verify again.** Generation also re-checks.
- Throttles stop SMS pumping: at most 5 sends per phone per hour, plus IP and device limits at the gateway.
- Only the HMAC of the code is stored, so a database leak cannot replay codes.
- Entering the OTP also records the party's **consent** (DPDP notice version) to processing their name and number for this document.
- The document records "Mobile +91******0002 verified via SMS OTP on IST" under the party's signature block.
- **Limitation:** OTP proves control of a phone number, **not identity**. Phase 2 adds DigiLocker/Aadhaar-based KYC for buyers.

4.3 Pincode → state → rate mapping logic



Duty bases:

- `FIXED` : a flat amount.

- `CONSIDERATION`: rate × consideration.
- `MAX_CONSIDERATION_MARKET_VALUE`: rate × max(consideration, circle-rate value), from the `circle_rates` table.
- `AVG_ANNUAL_RENT_PLUS_DEPOSIT`: for terms of 12 months or less, rate × (rent for the whole term + deposit); for longer terms, rate × (12 × rent + deposit).
- `TOTAL_RENT_PLUS_DEPOSIT`: rate × (rent × term + deposit).
- `MH_LEAVE_AND_LICENCE`: rate × (licence fee × term + 10% p.a. notional interest on the deposit).
- `SLAB`: bracket rates.

Unknown condition keys **never match**, so a rule the engine doesn't understand is never applied silently.

Worked example (illustrative seed, Gurugram 122001, rent ₹25,000 × 11 months, deposit ₹50,000):

Line	Amount
Stamp duty, HR, 1.5% × (₹2,75,000 + ₹50,000)	₹4,875
Service fee	₹199
E-stamp vendor charge	₹50
GST 18% on ₹249	₹44.82
Total	**₹5,168.82**

Rate maintenance:

- Rates are edited only in the ops console, with **maker-checker** approval.
- A new notification becomes a new row with `effective_from`; the old row gets an `effective_to`. The exclusion constraint rejects overlaps.
- A weekly job diffs state IGR pages and alerts legal ops to changes.

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5. (d) Template JSON Schema

There are two schemas:

- *`schemas/document-template.schema.json`* is the **template definition**: what a lawyer and product team author.
- *`schemas/document-instance.schema.json`* is the **filled document snapshot**: the frozen data an output was rendered from. It is stored with every generated file so it can be re-rendered byte for byte.

Template top-level blocks:

Block	Purpose
`jurisdiction`	Allowed and excluded states
`legalMeta`	Governing laws, `registration.mandatoryIf` (e.g. term > 11), `eSignPermitted`, `lawyerReviewRecommended`, reviewer

Block	Purpose
`parties[]`	Role, label, min/max, `otpRequired`, `autofillFromProfile`, per-party fields
`sections[].fields[]`	Dynamic form: field `type` (text, money, date, pincode, pan, aadhaarLast4, repeater …), `validation`, `showIf`, `autofillFrom`, `pii`
`pincodeBinding`	Which field holds the property pincode; what to derive (`state`, `district`, `sro`, `areaType`, `stampRates`, `registrationFee`); whether to lock to the jurisdiction
`stampDuty.inputs`	Maps form paths to engine inputs (`monthlyRentPaise → terms.monthlyRent`, …). **No rates here**
`eStamp`	`firstPartyRole`, `secondPartyRole`, `dutyPaidByRole`, `descriptionOfDocument`, per-state `articleCodes`, `fulfilmentModes`, `placement.footerText` (certificate number on every page)
`verification`	`requiredRoles`, `channels`, OTP length, TTL and attempts, `blockGenerationUntilVerified`, consent text version
`body`	Preamble, recitals, clauses (Handlebars text with optional `showIf` / `states`), schedules
`signatureBlocks[]`	Per role: photo box, thumb impression, "verified" badge
`output`	`formats` (PDF, DOCX), page size, margins, draft watermark, file name
`disclaimer`	Text, `requireAcknowledgement`

Instance `$defs`:

- **`partyVerification`**: `{required, status, channel, verifiedAt, verificationId, consentTextVersion}`. Conditional: when status is `VERIFIED`, `channel`, `verifiedAt` and `verificationId` are required.
- **`pincodeData`**: `{pincode, postOffice, stateCode, stateName, district, districtLgdCode, areaType, sro{code,name}, resolution{confidence, userConfirmed, candidates}}`
- **`dutyQuote`**: `{lines[{chargeType, amountPaise, basis, baseAmountPaise, rateBps, rateId, articleRef, explanation, payableAt}], totalGovtPaise, estampAmountPaise, quoteHash, verifiedRates}`
- **`eStamp`**: `{transactionId, status, provider, fulfilmentMode, certificateNo, uniqueDocReference, issuedAt, accCode, articleCode, descriptionOfDocument, propertyDescription, considerationPaise, stampDutyPaise, firstParty, secondParty, dutyPaidBy, purchasedBy, verificationUrl}`. Conditional: an issued status requires `certificateNo` and `issuedAt`.

An abridged excerpt from *`templates/rent-agreement.residential.json`*:

```
{
  "templateKey": "rent-agreement.residential",
  "version": "1.0.0",
  "documentType": "RENT_AGREEMENT",
  "jurisdiction": { "states": ["DL", "HR", "KA"], "excludedStates": ["MH"] },
  "legalMeta": {
    "registration": { "mandatory": false, "mandatoryIf": { "field": "terms.termMonths", "op":
"gt", "value": 11 } },
    "maxTermMonths": 11, "eSignPermitted": false, "lawyerReviewRecommended": true
  },
  "parties": [
    { "role": "LANDLORD", "min": 1, "max": 4, "otpRequired": false, "autofillFromProfile": true,
"fields": ["…"] },
    { "role": "TENANT",   "min": 1, "max": 4, "otpRequired": true,  "fields": ["…"] }
  ],
  "pincodeBinding": { "fieldPath": "property.pincode", "derive": ["state", "district", "sro",
"areaType", "stampRates", "registrationFee"] },
  "stampDuty": { "inputs": { "monthlyRentPaise": "terms.monthlyRent", "securityDepositPaise":
"terms.securityDeposit", "termMonths": "terms.termMonths" } },
  "eStamp": {
    "required": true, "firstPartyRole": "LANDLORD", "secondPartyRole": "TENANT",
"dutyPaidByRole": "TENANT",
    "descriptionOfDocument": "Rent Agreement",
    "articleCodes": { "DL": "Article 35 (a)(i) Lease", "HR": "Article 35 Lease", "KA": "Article
30 (a)(i) Lease", "*": "Lease" },
    "fulfilmentModes": ["DIGITAL_SELF_PRINT", "ACC_PRINT_COURIER"],
    "placement": { "certificateAsFirstPage": true, "footerText": "E-Stamp Certificate No.
{{eStamp.certificateNo}} | Ref {{referenceNo}} | Page {{page}} of {{pages}}" }
  },
  "verification": { "requiredRoles": ["TENANT"], "channels": ["SMS", "WHATSAPP"],
"otpTtlSeconds": 300, "maxAttempts": 5, "blockGenerationUntilVerified": true }
}
```

The test suite validates every template against the schema, and the filled snapshot against the instance schema. A sample snapshot is in `samples/rent-agreement-instance.json`.

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6. (e) Build plan

The team assumed for the MVP is 1 PM, 1 designer, 2 backend, 2 mobile, 1 QA, plus a part-time advocate and 1 ops agent.

MVP: 10–12 weeks · "Rent agreements in NCR and Bengaluru, stamped online"

Weeks	Deliverables
0–2 (in parallel, **start day 1**)	**Long-lead items:** e-stamp partner contract (SHCIL partner/ACC or aggregator), TRAI DLT registration (entity, header, templates), Razorpay KYC and the settlement structure for pass-through duty, WhatsApp Business verification, legal opinion on e-sign for leases, DPDP privacy notice, T&Cs and disclaimer

Weeks	Deliverables
1–3	Foundations: monorepo, CI, AWS Mumbai (Terraform), Postgres schema and migrations, phone-OTP login, profile, India Post pincode import, curated SRO map for DL/HR (Gurugram)/UP (Noida)/KA (Bengaluru)
3–5	Template engine (JSON Schema, dynamic forms, `showIf`, autofill), Rent Agreement + General Affidavit + NOC + Rent Receipt templates, advocate review
4–6	Party management and **tenant OTP** (MSG91 SMS + WhatsApp, fallback, throttles, DLR webhooks)
5–7	Duty engine + `stamp_rates` for 4 states (verified by the advocate), quote screen, Razorpay checkout and webhooks, refunds
6–9	**E-stamp:** adapter for the first partner, `MANUAL_OPS` fallback with an ops console (queue, upload, SLA timer), poller, reconciliation report
8–10	Output: HTML → PDF (Chromium), certificate merge, footer with certificate number and QR, DOCX, S3 with presigned links, public "verify this document" page
10–12	QA (end-to-end on real ₹10 test stamps), security review and VAPT, load test on render workers, store submission, soft launch

MVP exit criteria:

- Rent agreement from pincode to stamped PDF in under 10 minutes on the 80th percentile.
- OTP delivery above 97% within 30 seconds.
- E-stamp issued within 15 minutes on the 90th percentile for API states, and within 4 business hours for manual states.
- Zero double purchases.
- 100% of `stamp_rates` rows verified.

Phase 2: months 4–7 · "More documents, more states, trust"

- Documents: Maharashtra Leave & Licence with **IGR e-registration** hand-off (biometric appointment flow), Agreement to Sell (full), GPA/SPA, Indemnity Bond, Commercial Lease, Token Receipt, Possession Letter.
- States: MH, UP, TN, TS, GJ, WB (data plus partner coverage).
- Trust: DigiLocker and PAN verification for buyers and sellers. **Aadhaar eSign** for documents where it is legally permitted. A video-KYC option for high-value deals.
- **Advocate review marketplace**: paid review with 24-hour turnaround and comments on the draft.
- Doorstep delivery of stamp paper (courier), and notary partner booking.
- Regional-language templates (Hindi, Marathi, Kannada) with Noto fonts and a bilingual layout.
- Broker accounts: saved parties, bulk create, prepaid wallet, GST invoices.
- Renewal reminders (day 300 of 11 months) and one-tap renewal with re-stamping.

Phase 3: months 8–14 · "Registration-grade transactions"

- Sale deed, gift, release and partition deeds:
  - market-value check against `circle_rates`,
  - stamp duty on market value,

    - integrated **SRO appointment booking** and pre-registration data entry on the state IGR portals,
    - checklists (PAN or Form 60, TDS under s.194-IA, photographs).
- **Franking** route and high-value e-stamp (above ₹1 lakh duty) with escrow-style holding.
- Home-loan partner APIs: banks request a mortgage deed or MODT pack.
- Wills (no stamp duty), with optional registration and witness video.
- B2B API and white-label for property portals and banks.
- ML assist:
    - flag inconsistent inputs (rent far above the locality median, deposit greater than 10× rent),
    - clause suggestions from the advocate-curated library only. No free-form generated legal text in documents.
- Fraud graph: the same phone or PAN across many properties triggers ops review.

---

7. (f) Sample code

`server/` is a runnable reference implementation of the Rent Agreement flow with **OTP verification and e-stamp purchase**. It runs with no external accounts (sandbox providers) and switches to the real MSG91, WhatsApp, Razorpay and aggregator adapters when their environment variables are set.

File	What it shows
`src/pincode/pincodeService.ts`	Pincode → state, districts, area type, SRO candidates; auto-select versus confirmation
`src/stamp/dutyEngine.ts`	Rate selection by specificity, all duty bases, min/max/rounding, quote hash
`src/otp/otpService.ts`	CSPRNG OTP, HMAC storage, TTL, attempts and lock, resend cooldown, per-phone cap, WhatsApp → SMS fallback, phone-change guard
`src/otp/providers.ts`	MSG91 Flow API (DLT), Meta WhatsApp Cloud authentication template, console provider
`src/estamp/provider.ts`, `providers.ts`	Vendor interface; SHCIL sandbox (renders a mock certificate); generic HMAC-signed aggregator REST adapter with Idempotency-Key; manual-ops queue
`src/estamp/estampService.ts`	Paid-and-quote-matched check, idempotency, ambiguous-error handling, polling to manual review, ops completion
`src/payments/razorpay.ts`	Orders API, checkout signature `HMAC(order_id
`src/documents/template.ts`	Loads and validates a template against the JSON Schema; form validation; `showIf` evaluator
`src/documents/render.ts`, `output.ts`	Handlebars (amounts in words in the Indian system, dates as "1st day of November 2026"), HTML → PDF via Chromium, certificate page + footer + QR via pdf-lib, DOCX
`src/rentAgreement/rentAgreementService.ts`	**The orchestrator**: a state-guarded flow from draft to generated

File	What it shows
`src/http/app.ts`	Fastify routes (§9) with zod validation and error mapping
`test/flow.test.ts`	End to end over HTTP: OTP gate blocks the quote → wrong then right OTP → quote ₹4,875 → bad signature rejected → paid → certificate `IN-HR…` → webhook replay doesn't double-buy → PDF and DOCX → snapshot validates against the instance schema → another user is denied
`mobile/RentAgreementCheckout.tsx`	React Native: OTP card (SMS/WhatsApp, autofill, cooldown, attempts left) and checkout (duty breakdown, disclaimer acknowledgement, Razorpay, download and share)

To run:

```
cd legalkart/server
npm install
npm test            # 13 tests: duty engine, pincode, OTP, template schema, full HTTP flow
npm run demo        # writes ../samples/rent-agreement-sample.{pdf,docx,html} + instance JSON
npm run dev         # API on :8080 (sandbox providers; OTPs printed to console)
```

PDF output needs Chromium. Set `CHROMIUM_PATH` (defaults to `/opt/pw-browsers/chromium`). Production environment variables:

- **Auth:** `OTP_PEPPER`
- **SMS:** `MSG91_AUTH_KEY`, `MSG91_FLOW_TEMPLATE_ID`, `DLT_TEMPLATE_ID`, `ANDROID_SMS_APP_HASH`
- **WhatsApp:** `WA_ACCESS_TOKEN`, `WA_PHONE_NUMBER_ID`, `WA_OTP_TEMPLATE`
- **Razorpay:** `RAZORPAY_KEY_ID`, `RAZORPAY_KEY_SECRET`, `RAZORPAY_WEBHOOK_SECRET`
- **E-stamp aggregator:** `ESTAMP_AGGREGATOR_URL`, `ESTAMP_AGGREGATOR_KEY`, `ESTAMP_AGGREGATOR_SECRET`, `ESTAMP_AGGREGATOR_STATES`
- **Pricing and links:** `SERVICE_FEE_PAISE`, `ESTAMP_VENDOR_FEE_PAISE`, `VERIFY_BASE_URL`

What remains production work (deliberately out of the sample):

- Postgres repositories (the in-memory `MemoryStore` mirrors `schema.sql`)
- BullMQ jobs (the sample buys inline)
- JWT auth (the sample uses an `x-user-id` stub)
- S3 storage and KMS field encryption
- `@fastify/rate-limit`
- The real vendor contract mapping inside `AggregatorHttpProvider`

---

8. Compliance, legal & risk register

Area	Requirement / risk	Mitigation
Not a law firm	Bar Council rules restrict advocates' advertising and solicitation. A platform must not hold itself out as giving legal advice	Clear disclaimer with acknowledgement; templates reviewed by an enrolled advocate (`reviewed_by`); advocate review sold as a referral to independent advocates

Area	Requirement / risk	Mitigation
Stamp law	Under-stamped instruments are inadmissible until duty and penalty are paid (Indian Stamp Act / state Acts)	Data-driven rates with provenance; verified-only in production; the quote shows the article and basis; the user can't override the duty
Registration	Leases over 1 year, and conveyances, must be registered (Registration Act s.17). Maharashtra L&L always	Template `mandatoryIf` blocks term > 11 for rent; registration-mandatory documents route to the SRO flow (Phase 2/3)
E-signatures	IT Act First Schedule excludes POA, will, trust, and contracts for sale or conveyance of immovable property	`eSignPermitted: false` for those; wet-ink signature blocks; counsel opinion before enabling e-sign for leases
Data protection	DPDP Act 2023 and Rules: notice, consent, purpose limitation, retention, breach reporting, Data Principal rights	Consent log per party (OTP consent text version); India-only storage; retention policy (e.g. drafts 90 days, issued documents 8 years or as the user chooses); deletion and export APIs
Aadhaar	Aadhaar Act and UIDAI rules: no storage of full Aadhaar numbers by non-authorised entities	Only the last 4 digits; KYC through licensed KUAs/ESPs only
SMS	TRAI TCCCPR / DLT; content must match the registered template	Templates locked in config; DLR monitoring; header registered
Payments	RBI PA guidelines; pass-through government money; GST on services	Separate ledger lines; GST invoices; Razorpay settlement structure agreed with the PA and an accountant
Fraud	Fake landlord or tenant identities; one property rented to many	OTP on tenants/buyers (MVP); PAN/DigiLocker (P2); cross-document anomaly checks (P3); verification QR on every page
Vendor dependence	E-stamp partner downtime	Multiple adapters; manual-ops fallback; status page; automatic refund on failure
Rate drift	A state revises duty	Effective-dated rows; weekly diff job; quote frozen with a hash; re-quote if the start date crosses `effective_to`

9. API reference

All routes are JSON and authenticated with `Authorization: Bearer <jwt>`. The sample uses an `x-user-id` stub instead. Money is in paise.

Method	Path	Purpose
GET	`/v1/pincodes/:pincode`	Resolve state, districts, area type, SRO candidates, `needsUserConfirmation`

Method	Path	Purpose
GET	`/v1/templates/rent-agreement`	Template JSON (the app renders forms from it)
POST	`/v1/rent-agreements` `{pincode, sroCode?}`	Create a draft and freeze the pincode jurisdiction
PUT	`/v1/rent-agreements/:id/sro` `{sroCode}`	Confirm the SRO when ambiguous
PUT	`/v1/rent-agreements/:id/form`	Save answers → `{errors[]}`
POST	`/v1/rent-agreements/:id/parties` `{role, fullName, phoneE164, details}`	Add a party (`otpRequired` set from the template)
PATCH	`/v1/rent-agreements/:id/parties/:partyId` `{phoneE164}`	Change the phone (resets verification)
POST	`/v1/rent-agreements/:id/parties/:partyId/otp` `{channel}`	Send an OTP (SMS / WhatsApp)
POST	`/v1/rent-agreements/:id/verifications/:verificationId` `{otp}`	Verify → `{pendingPartyIds}`
POST	`/v1/rent-agreements/:id/quote`	Duty lines + fees + GST. **409** **`VERIFICATION_PENDING`** until tenants verify
GET	`/v1/rent-agreements/:id/preview`	Watermarked HTML draft
POST	`/v1/rent-agreements/:id/payments`	Create a Razorpay order for the frozen quote
POST	`/v1/rent-agreements/:id/payments/confirm`	Verify the checkout signature → capture → buy the e-stamp
POST	`/v1/webhooks/razorpay`	`payment.captured` (HMAC on the raw body, deduped)
POST	`/v1/rent-agreements/:id/estamp` `{fulfilmentMode}`	(Re)trigger the e-stamp purchase. Idempotent
POST	`/v1/rent-agreements/:id/generate`	Stamped PDF + DOCX → file links and SHA-256
GET	`/v1/rent-agreements/:id/files/:format`	Download `pdf` or `docx`

Error shape: `{code, message, details?}` . Codes:

- `VALIDATION` , `FORM_INCOMPLETE` , `PARTY_MISSING`
- `VERIFICATION_PENDING` , `INVALID_OTP` (`details.attemptsLeft`), `LOCKED` , `EXPIRED` , `RESEND_TOO_SOON` , `RATE_LIMITED`
- `STATE_NOT_SUPPORTED` , `NO_RATE` , `UNVERIFIED_RATE`
- `BAD_SIGNATURE` , `NOT_PAID` , `AMOUNT_MISMATCH` , `NOT_STAMPED` , `INVALID_STATE`